# A Head Camera Is (Almost) All You Need

An electronics-free UMI glove labelled and observed by one moving camera

Yosub Shin[1] Yuan Liu[2] Igor Molybog[1] [1]University of Hawai'i at Mānoa [2]Independent

arXiv:2610.27622
The UMI Arena, CoRL 2026

## Why a head camera only

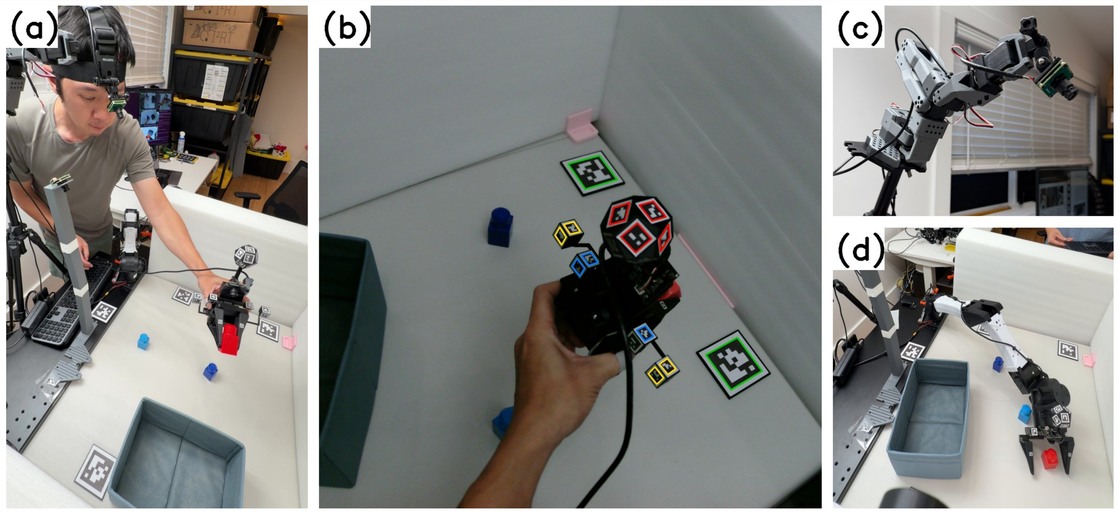
- Wrist devices need a **second stream** to track the gripper: SLAM, an IMU, a controller or an external camera.
- **Gaze leads the hand**: the gripper stays in view, at arm's length, with the goal.
- The frames the policy sees are the frames that **carry the labels**.

No SLAM, no IMU, no wrist camera, no cables.

**$144** glove, camera, strap

**103 g** on the head

## One head camera: the policy's observation and the pose labels




(a) A $122 4K head camera, a $22 electronics-free glove, four paper tiles. (b) The camera's own frame: tiles give the camera pose, the ball the gripper pose, the jaw tags the opening. (c) At deployment the same camera rides a $122 six-joint arm (d) beside the robot.

## Label accuracy

- **Camera pose** from the paper tiles, refit from the footage to 0.5 px.
- **Gripper pose** by PnP on a 15 mm ArUco dodecahedron.
- **Jaw width** from printed tails on the jaws.

**6.5 mm** tool point — **1.6°** orientation — **3.5 %** jaw width, of stroke

Median error against the robot's encoders; a label on 93–100 % of frames.

## Collection

**2–3×** demos per minute vs teleop — **198** demos in four sessions

**6–8 s** per demo; teleop 14 s — **0** robot time

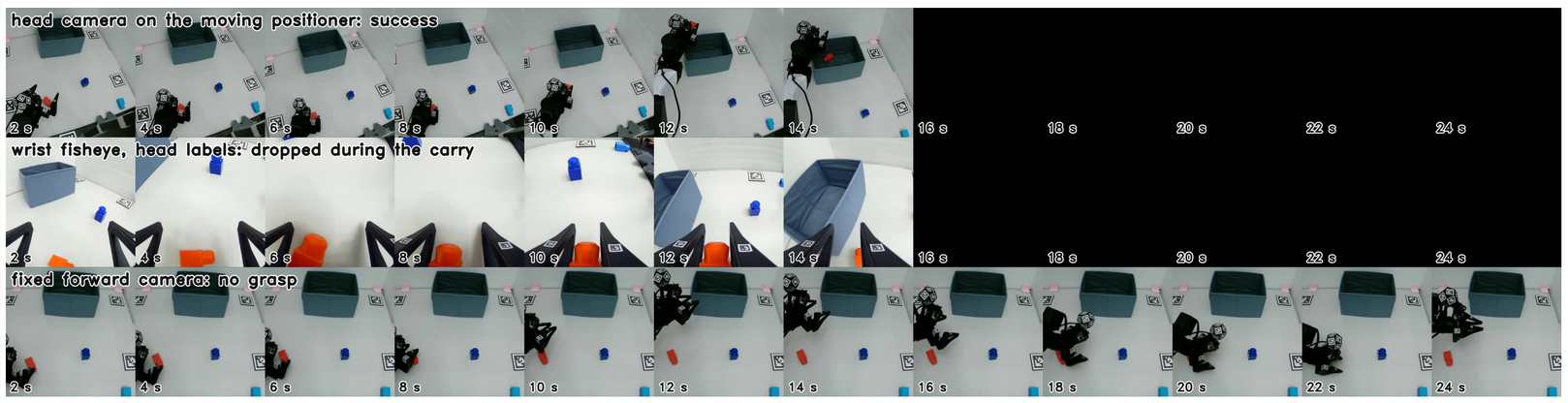
Continuous recording, episodes cut by a keypress; 0.7–7.4 % of frames dropped for tracking.

## Deployment

- The same camera on a $122 six-joint arm beside the YAM.
- It follows the gripper **from the robot's encoders alone**, in teleoperation and in rollouts, so the policy sees what the collector saw.

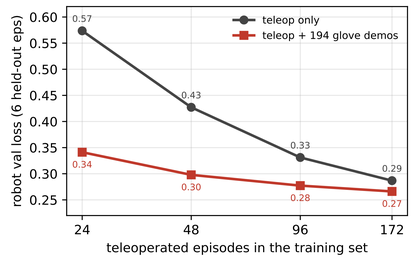
**46 mm** head pose vs the collector — **5.4°** static camera: 87 mm, 7.0°

## One placement, seen by each policy




Placement 23, each policy's own frames every 2 s on one clock, black once its episode ended. Head camera: gripper and bin in one frame through the carry, done in 16 s. Wrist fisheye: only block and wall after the grasp; drops the block before reaching the bin. Fixed camera: sees the whole scene but never closes the gripper (41 s, cut here).

## Cheap glove demos complement a small teleop set




**10/25**
48 teleop episodes

**17/25**
48 teleop + 194 glove demos

**1 ≈ 2–3**
teleop episode ≈ glove demos

9 placements won to 2 lost ($p = 0.065$). On the curve (head-label chain, own held-out targets) the 194 glove demos are worth **65–100 teleop episodes**. With all 172 they still help: 18/25 → 20/25 (4 won to 2 lost, $p = 0.69$).

## Head-camera labels can replace the encoders

| teleop labelled by | encoders | head camera |
|---|---|---|
| **validation loss** | 0.318 | 0.328 (+3 %) |
| **rollouts** | 18/25 | 15/25 ($p = 0.58$) |

Same 172 teleop episodes, same held-out encoder targets; 5 placements won to 8 lost ($p = 0.58$).

## Where each view fails

| failures of 25 | no grasp | **dropped** | success |
|---|---|---|---|
| head camera | 5 | **0** | 20 |
| wrist fisheye (UMI) | 6 | **4** | 15 |
| wrist fisheye, head labels | 3 | **8** | 14 |
| fixed camera | 12 | **1** | 12 |

Dropped = during the carry or at the bin. All four co-trained; offline loss within 0.011 of each other.

Wrist-camera policies **drop the block mid-carry**, every time while the fisheye looked away from the bin. The head camera and the fixed camera keep the bin in frame and almost never drop; they fail by not grasping. Caveat: the teleoperator watched the head-camera preview while demonstrating.

## It is the view, not the labels

- Same wrist view, labels from the fixed camera or from the head camera: **15/25 vs 14/25**, no change.
- Same head labels, wrist view or head view: **14/25 → 20/25**, and the drops disappear.

## Limits and next

One task, one collector, one tile layout; teleoperation watched the head-camera view. Glove-only policies do not transfer: co-training is required. Next: bimanual, tile-free in the wild.

Setup: eight ACT policies, joint-space actions. Rollouts: 25 placements, one trial each, 60 s. Every co-trained mix uses the same 194 glove demos.